

Computer Networks — Unit Test 1

Subject Code: CN301 | Duration: 30 Minutes | Total Marks: 10

Instructions: This is a Multiple Choice Question paper. Each question carries 1 mark. There is no negative marking. Choose the single best answer for each question.

Q1. Which layer of the OSI model is responsible for routing packets between different networks?

- A) Data Link Layer
- B) Network Layer
- C) Transport Layer
- D) Session Layer

Q2. What does DNS stand for?

- A) Domain Name System
- B) Dynamic Network Service
- C) Data Naming Standard
- D) Distributed Node Server

Q3. Which protocol is used to send email between mail servers?

- A) IMAP
- B) POP3
- C) SMTP
- D) FTP

Q4. In TCP, which mechanism is used to avoid overwhelming a slow receiver?

- A) Flow Control
- B) Congestion Avoidance
- C) Error Correction
- D) Multiplexing

Q5. What is the default subnet mask for a Class C IP address?

- A) 255.0.0.0
- B) 255.255.0.0
- C) 255.255.255.0
- D) 255.255.255.255

Q6. Which of the following is a connectionless transport-layer protocol?

- A) TCP
- B) UDP
- C) HTTP
- D) FTP

Q7. Which device operates primarily at the Data Link layer and forwards frames based on MAC addresses?

- A) Router
- B) Hub
- C) Switch
- D) Repeater

Q8. What is the primary purpose of the ARP protocol?

- A) Translate domain names to IP addresses
- B) Translate IP addresses to MAC addresses
- C) Encrypt network traffic
- D) Assign IP addresses dynamically

Q9. Which port number is used by HTTPS by default?

- A) 21
- B) 80
- C) 443
- D) 25

Q10. Which of these best describes a firewall's function?

- A) Speeds up DNS resolution
- B) Filters network traffic based on security rules
- C) Compresses data before transmission
- D) Assigns bandwidth priority to applications